

Course Contents		
UNIT-I	Introduction to Generative AI and ChatGPT	8 Hours
Understanding Generative AI: Concepts and applications. Evolution of Language Models: From rule-based systems to large language models. Introduction to ChatGPT: Features, capabilities, and use cases. Navigating Ethical Considerations: AI-generated content and responsible usage		
UNIT-II	Prompt Engineering and Basic Techniques	8 Hours
Basics of Prompt Engineering: Crafting effective prompts for desired outputs. Generating Text with ChatGPT: Single-sentence completions and short responses. Fine-tuning Concepts: Overview of fine-tuning and transfer learning Workshop: Creating basic prompts for specific tasks		
UNIT-III	Advanced Prompting Strategies	8 Hours
Chain-of-Thought Prompting: Generating coherent multi-turn responses. Zero- and Few-shot Learning: Leveraging model's general knowledge and adapting to specific tasks. Prompt Injunctions: Controlling model behavior through instructions. Prompt Parameter Tuning: Adjusting parameters to influence output style. Workshop: Applying advanced techniques to real-world scenarios.		
UNIT-IV	Multimodal Generative AI	8 Hours
Introduction to Multimodal Models: Combining text and images for richer outputs. Stable Diffusion and Mid Journey Models: Generating images and text together. Use Cases in Creative Industries: Design, photography, and multimedia content generation. Workshop: Creating prompts for multimodal outputs		
UNIT-V	Applications in Code Generation and Data Science	7 Hours
Code Generation with LLMs: Writing code snippets for various programming tasks. Data Science Applications: Using LLMs for data preprocessing, analysis, and modeling. Introduction to Copilot: Collaborative coding with AI assistance. Workshop: Generating code and conducting data tasks with AI.		
UNIT-VI	Applications in Code Generation and Data Science	6 Hours
Product Development Fundamentals: Design thinking, user experience, and iteration. Integrating AI in Applications: Incorporating ChatGPT or similar models. Web Application Development with Flask: Creating a GPT-powered web app. Final Projects: Students develop their own AI-enabled applications		
TEXTBOOKS: - Generative Deep Learning. - Deep Learning for Natural Language Processing - Hands-On Generative AI with Transformers and Diffusion Models - Machine Learning with PyTorch and Scikit-Learn		